

04-638: Programming for Data Analytics

Analytics Assignment II: Predictive Modeling and Hypothesis Testing

[Group Assignment - Extension of EDA Assignment]

Release Date: 15th November, 2025

Due Date: 24th November, 2025 - 10:00 am CAT

Points Total: 100 pts

Submission Requirements

Submit the a zip file on Canvas containing:

1. **Python Scripts (.py files)** - Modularized code following OOP principles
2. **Jupyter Notebook** - Main analysis workflow that imports and uses your modules
3. **PDF Report** (Max: 2000 words)

Assignment Context

Building upon your exploratory data analysis from Assignment I, you will now test hypotheses and make predictions using supervised machine learning models. This assignment emphasizes how to:

1. Formulate testable hypotheses from exploratory data analysis
2. Design and implement supervised learning models for prediction
3. Apply OOP principles to create modular, reusable code
4. Evaluate and compare multiple machine learning models
5. Justify modeling decisions based on data characteristics
6. Communicate actionable insights to concerned stakeholders

Task Description

Part 1: Hypothesis Formulation (10 points)

Based on your EDA findings from Assignment I:

[5 pts] Formulate **three (3) clear, testable hypotheses** about your dataset. Each hypothesis should:

- Be specific and measurable
- Relate to patterns or relationships discovered in your EDA
- Be testable using supervised learning approaches
- Include a null hypothesis and alternative hypothesis

[5 pts] For each hypothesis:

- Explain the rationale (why this hypothesis matters)
- Define what would constitute evidence to accept or reject the hypothesis
- Identify the target variable(s) for prediction
- Specify whether this is a classification or regression problem

Part 2: Data Preparation and Feature Engineering (15 points)

[5 pts] Data Cleaning:

- Handle missing values with justified strategies
- Address outliers discovered in EDA (removal, transformation, or retention with justification)
- Ensure data quality and consistency

[5 pts] Feature Engineering:

- Create new features that may improve model performance
- Transform features as needed (scaling, encoding, normalization)
- Document the reasoning behind each engineered feature

[5 pts] Train-Test Split:

- Properly split data into training and testing sets
- Justify your split ratio
- Address any class imbalance issues (if applicable)
- Ensure no data leakage

Part 3: Model Development and Implementation (35 points)

[15 pts] Model Selection and Implementation:

Implement **at least three (3) different supervised learning models** appropriate for your problem:

For each model:

- Implement using scikit-learn or similar libraries
- Perform hyperparameter tuning (GridSearchCV, RandomizedSearchCV, or manual tuning)
- Document hyperparameters tested and final selections

[10 pts] Code Modularization (OOP Implementation):

Create well-structured Python modules using OOP principles:

Required modules/classes:

- **DataProcessor class**: Handles data loading, cleaning, and preprocessing
- **FeatureEngineer class**: Manages feature creation and transformation
- **ModelTrainer class**: Encapsulates model training and hyperparameter tuning
- **ModelEvaluator class**: Handles model evaluation and comparison

Each class should:

- Follow single responsibility principle
- Include proper docstrings
- Have clear method names and functionality
- Use appropriate encapsulation

[10 pts] Model Justification:

- Explain why each model was chosen for your specific problem
- Discuss the strengths and limitations of each approach
- Connect model choice to data characteristics (size, linearity, feature relationships, etc.)

Part 4: Model Evaluation and Comparison (20 points)

[10 pts] Performance Metrics:

Use appropriate metrics for your problem type:

Evaluate all models using:

- Training set performance
- Testing set performance
- Cross-validation scores (5-fold or 10-fold)

[5 pts] Model Comparison:

- Create comparison tables/visualizations showing model performance
- Identify the best-performing model with justification
- Discuss overfitting/underfitting issues observed

[5 pts] Feature Importance:

- Analyze and visualize feature importance for applicable models
- Discuss which features most influence predictions
- Connect findings back to EDA insights

Part 5: Hypothesis Testing Results (10 points)

[6 pts] For each of your three hypotheses:

- State whether you accept or reject the hypothesis
- Provide quantitative evidence from model results
- Discuss the strength of the evidence
- Note any limitations or caveats

[4 pts] Discuss insights gained:

- What did the predictive models reveal that EDA did not?
- Were there any surprising findings?
- How confident are you in the conclusions?

Part 6: Code Quality and Documentation (10 points)

[4 pts] Code Organization:

- Clear directory structure
- Logical separation of modules
- Main notebook that cleanly imports and uses modules

[3 pts] Documentation:

- Comprehensive docstrings for all classes and methods
- Inline comments for complex logic
- README file explaining how to run the code

[3 pts] Code Quality:

- Follows PEP 8 style guidelines
- No code duplication
- Efficient implementations
- Proper error handling

Deliverables Breakdown

1. Python Scripts (Modularized Code)

Your code should be organized as follows:

Python

```
project_directory/
|
├── src/
|   ├── __init__.py
|   ├── data_processor.py      # DataProcessor class
|   ├── feature_engineer.py   # FeatureEngineer class
|   ├── model_trainer.py      # ModelTrainer class
|   ├── model_evaluator.py    # ModelEvaluator class
|   └── utils.py              # Helper functions
|
├── notebooks/
|   └── main_analysis.ipynb    # Main workflow notebook
|
├── results/
|   ├── figures/               # Generated plots
|   └── models/                # Saved model files
|
├── README.md
└── requirements.txt
```

Note: You do not need to submit your dataset

2. Jupyter Notebook

- Import and demonstrate use of your custom modules
- Show complete workflow from data loading to final predictions
- Include markdown cells explaining each major step
- Display key visualizations and results

3. PDF Report (Max: 2000 words)

Required Sections:

1. **Abstract** (200 words)
2. **Introduction and Hypotheses**
 - Background from EDA
 - Three hypotheses with rationale
3. **Methodology**
 - Data preparation approach

- Feature engineering decisions
- Model selection rationale
- Evaluation strategy
- 4. **Results**
 - Model performance comparison
 - Feature importance findings
 - Hypothesis testing outcomes
- 5. **Discussion**
 - Interpretation of results
 - Comparison with EDA insights
 - Limitations and assumptions
- 6. **Conclusion**
 - Summary of findings
 - Practical implications
 - Future work recommendations
- 7. **References** (minimum 6)